

Divyesh Sankhla — Master Content Repository (Full, Detailed)

Use this as your single source of truth. Copy/paste and prune per JD. Bullets include variants with metrics, tech stacks, and outcomes so you can tailor quickly.

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Professional Summary (Long Form)

Embedded/Edge Software Engineer with 3+ years building Linux-based camera firmware and low-latency streaming systems at scale. Specializes in ONVIF-compliant IP cameras, OTA update pipelines, secure media transport (RTSP/RTSPS/WebRTC), and resilient cloud recording services with TLS/digest authentication. Comfortable across the stack: C/C++ on embedded Linux; multithreading, queues, and lock management; ROS-based real-time robotics; GPU acceleration with CUDA; and protocol work from UART/SPI/I2C up to MQTT over TLS. Proven in cross-company collaborations (Honeywell, Hanwha), integrating 15+ APIs and shipping features across 4–5+ SKUs. Strong fundamentals in OS (scheduling, synchronization, interrupts) via Pintos kernel projects. Experience extends to data analysis/ML for behavioral insights. Known for deep debugging (gdb/Valgrind/Wireshark), refactoring large legacy codebases, and delivering reliable features under shifting product constraints.

Education

University at Buffalo (SUNY) — Buffalo, NY

Master of Science, Computer Science & Engineering — 2025–Present

Nirma University — Ahmedabad, India

B.Tech., Electronics & Communication Engineering — 07/2017–06/2021

- SPI: 7.33 | PPI: 7.2
 - Relevant Coursework: Data Communication & Networking, Embedded Systems, Wireless Sensor Networks, Data Structures, Modern Processor Architecture
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Core Skills & Technologies (Expanded)

- **Languages:** C, C++ (OOP, STL, concurrency), Python, Shell/Bash, Makefiles, SQL (SQLite3)
- **Embedded/Linux:** Linux user space, POSIX threads, signals, IPC; init scripts; cross-compilation; FreeRTOS; Ubuntu Core
- **Streaming & Media:** RTSP/RTSPS, RTP, WebRTC (libdatachannel), Live555, FFmpeg; jitter/buffer control; latency tuning
- **Network & Cloud:** TCP/IP, UDP, NAT, DHCP, STUN/TURN, HTTP/HTTPS, TLS/digest auth, MQTT/Mosquitto; WebSockets
- **ONVIF & Web:** ONVIF G/S/T profiles, gSOAP, Lighttpd; REST/SOAP API design; JSON/XML tooling (cJSON/libxml2)
- **Protocols (Bus/Device):** SPI, I2C, UART, RS485; GPIO; camera sensor interfaces (concepts)
- **Debug/DevOps:** gdb, Valgrind, perf, Wireshark, tcpdump, Jenkins, Docker, Postman, MQTTX, Git, JIRA
- **Hardware/Platforms:** AMB_S5L, Ambarella CV22/CV2, NVIDIA Jetson TX2, RPi, STM32/ARMv7/v8, ESP8266, 8085/8086, FPGA (Quartus)
- **Security:** TLS (OpenSSL), digest authentication, credential/entitlement flows
- **Robotics/Perception:** ROS (pub/sub, nodes), OpenCV (CV pipelines), CUDA (GPU acceleration), PID control

- **Data/ML (foundational):** preprocessing, EDA; models—XGBoost, Decision Trees, SVM, MLP, K-Means; basic model eval
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Professional Experience

Einfochips (An Arrow Company) — Ahmedabad, India

Embedded Engineer (Full-time) | Aug 2021 – Dec 2024

Scope: Linux firmware and edge applications for intelligent IP cameras across 5+ SKUs (Honeywell, Hanwha). Shipping features spanning ONVIF server/APIs, OTA firmware delivery, low-latency media streaming, secure cloud video recording, and entitlement/configuration systems.

1) Reusable Camera Firmware (Honeywell, multi-SKU)

- Implemented **ONVIF Server Profile APIs** (G/S/T) with **gSOAP + Lighttpd**, enabling interop with ONVIF NVR/VMS; hardened for a **1M+ user database** using SQLite3 (schema tuning, indices, WAL mode) for high-QPS reads.
- Enhanced **15+ ONVIF APIs**, resolving interop gaps and compliance edge cases observed in field testing; systematic compatibility testing against multiple NVR/VMS suites.
- Built **WebSocket access** for tunneled HTTP requests to device services (auth, rate limiting, error codes), reducing support incidents for remote control.
- Led **repository refactoring** across modules (C/C++), eliminating duplicated utilities and consolidating XML/JSON parsers → improved build times and reduced defect surface area across a 7-person team.
- Created **reusable XML/JSON utility functions** (cJSON/libxml2) for device configuration, cutting boilerplate and decreasing integration time for 9+ team members.

Selected Outcomes: Fewer interop defects during certification; reduced config parsing defects; measurable drop in field issues related to malformed XML/JSON and ONVIF quirks.

2) OTA Firmware & Device Management

- Integrated **OTA update flow** into Linux camera firmware with integrity checks, staged downloads, version gating, rollback strategy, and post-update health checks.

- Supported **1,000+ devices** in the field (multi-SKU), designing update windows and bandwidth-aware chunking to minimize operational impact.
- Upgraded **SQLite3** schema and migrations around OTA/version metadata; added durable logging for auditability and recovery after unexpected power loss.

Selected Outcomes: Safe fleet-wide rollouts; reduced bricking risk; faster time-to-recover on failed updates via automated rollback.

3) Low-Latency Streaming & Real-time Media

- Designed a **Live555-based RTSP pipeline** and a **WebRTC streaming path** (libdatachannel) with Mosquitto for signaling/control; structured around **multithreaded C++** with lock-aware queues and linked lists for peer/message processing.
- Tuned **buffer sizes, RTP pacing, and jitter control** for cross-region streaming; integrated **TLS/digest authentication** for secure transport.
- Implemented **monitoring hooks** (stats/metrics) for RTT, packet loss, and end-to-end latency; provided tooling for Wireshark/tcpdump-based field diagnostics.

Selected Outcomes: Noticeable latency reduction for global viewers; improved reliability under lossy networks; faster root cause analysis via standardized metrics.

4) Cloud Video Recorder (CVR) & Resilience

- Engineered **RTSPS push** from edge to cloud recorder using C++ classes and threads; **event-based recording** with pre/post buffers; **30-second local fallback buffer** to bridge transient network failures.
- Added **digest/TLS** auth and token refresh; implemented **backpressure** and retry with exponential backoff.
- Built **event indexing** for quick lookup and retention policies (size- and time-based pruning), with durable metadata writes.

Selected Outcomes: Higher event capture reliability; graceful behavior during WAN outages; reduced data loss and support tickets.

5) Event Intelligence, Entitlements & Centralized Config

- Wrote an **Event Manager** that normalizes device-side events (motion, analytics) and publishes to cloud; enabled **NLP-based event search** downstream.

- Implemented an **Entitlement Manager** enforcing access control across **5+ feature domains** (live view, playback, administration, analytics, export) with role-based rules.
- Built **MQTT-based centralized settings** (Mosquitto) to fan out configuration to **4+ SKUs** with schema validation and transactional apply/rollback on failure.

Selected Outcomes: Faster fleet config changes; reduced misconfiguration risk; consistent RBAC enforcement across product lines.

Awards: Power Trainee & Best Team Award (2022, 2024 Annual Award Ceremonies).

Environment: C/C++, Linux, gSOAP/Lighttpd, SQLite3, Live555, WebRTC (libdatachannel), Mosquitto/MQTT, OpenSSL, cJSON/libxml2, FFmpeg; gdb, Valgrind, Jenkins, Docker, Wireshark, tcpdump, Git, JIRA.

Internship

Krossmark Innovations — Ahmedabad, India | Jun 2020 – Jul 2020

Sensor Data Monitoring (RS485/UART)

- Implemented **RS485** and **UART** communication between sensors and a centralized monitoring station across long-range links.
 - Added **packet framing, CRC, and re-try logic** to reduce dropped readings; basic buffering to smooth bursty traffic.
 - Integrated sensor drivers and validated data integrity via test harnesses.
 - **Stack:** C, UART/RS485 drivers, oscilloscopes/logic analyzers, Linux serial tools.
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Academic & Personal Projects (Deep Dives)

Autonomous Underwater Vehicle (AUV) — NVIDIA Jetson TX2 (2018–2020)

- **Objective:** Fully autonomous navigation and task completion under water using sensor fusion and CV pipelines.
- **Architecture:** ROS-based pub/sub messaging; camera + IMU/pressure sensors interfaced via microcontroller to Jetson TX2.

- **CV Pipeline:** OpenCV color/shape detection, morphological ops, contour analysis; tuned thresholds for underwater lighting. Achieved **~90%** detection accuracy in test conditions.
- **Control:** PID loops for thruster control; calibrated gains for stable heading/depth; failsafe states for sensor dropouts.
- **Performance:** CUDA-parallelized portions of image pipeline (e.g., filtering/convolution) achieving up to **~15×** speedups vs. CPU-only implementations (measured on representative frames).
- **Systems:** Battery management circuit for thrusters; watchdog processes; logging and replay tooling for post-mission analysis.
- **Role:** Software department lead; ROS node orchestration; CV & control integration; test planning on bench tank.
- **Stack:** ROS, C++/Python, OpenCV, CUDA, Jetson TX2, PID control, Linux.

Navigation Using Drone — Raspberry Pi + CV (2019)

- **Objective:** Real-time image-based navigation; compute on-board with RPi.
- **Work:** ROS nodes for frame capture → CV processing → direction messages; PID tuning for roll/pitch stabilization; safety interlocks.
- **Stack:** OpenCV, ROS, Python/C++, RPi GPIO/IMU integration.

Text Chat Application — Socket Programming (Single-threaded server)

- **Features:** User login/registration, direct messages, broadcast, blocking, offline buffering; CLI shell with commands.
- **Engineering:** **select()**-based I/O multiplexing; message framing; client state machines; durable queues for buffered delivery; simple auth.
- **Stack:** C/C++ (or Python variant if used), TCP sockets, POSIX APIs.

Operating Systems — Pintos Kernel Projects

- **Project 1: Threads & Scheduling**
 - Implemented **alarm clock** (tick-based sleep/wakeup via ordered sleep list).

- **Priority scheduling** with **priority donation** to address priority inversion.
- **Advanced scheduler** using **recent CPU** and **load average**; maintained ready queues; careful **interrupt disable/enable** windows for atomicity.
- **Skills**: Kernel C, synchronization primitives (locks, semaphores, cond vars), timer interrupts, list ordering, race detection.
- **Project 2: User Programs**
 - Implemented **system call handlers**: file I/O (open/read/write/close), process control (exec/wait/exit).
 - **Argument passing** into the user stack; validated user pointers; ensured **write protection** of executables.
 - Coordinated **parent/child synchronization** for process startup/failure signaling.
 - **Skills**: Process management, user/kernel memory isolation, file system integration, low-level debugging with gdb.

Digital Well-being Data Analysis (300K+ records)

- **Goal**: Analyze relationships between screen time, sleep quality, fatigue, and loneliness; identify mitigating factors (e.g., exercise).
- **Work**: Cleaning (missing/invalid handling), feature engineering, EDA visualizations; trained models: **XGBoost, Decision Trees, SVM, MLP, Logistic Regression, Stacking, K-Means** clustering.
- **Findings**: Prolonged screen time correlates with poorer sleep/health; **physical exercise** moderates adverse effects (directionally).
- **Stack**: Python, pandas, scikit-learn, matplotlib; basic cross-validation.

Digital Design & Hardware

- **16-bit RISC Processor (Verilog on Quartus)**: ISA design, datapath/control, ALU/regs/memory; test benches and waveform debugging.
- **FIFO Memory (Verilog)**: Parameterized depth/width; full/empty flags; timing validation and simulation.

- **Person Counting Device:** Two IR sensors at a single doorway; edge detection to infer direction (entry vs exit); simple debounce logic.
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Publications

- *Industrial Automation Using PLC, and IoT Advancements*, IJERT, 2021 — Manuscript accepted.
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Volunteer

- Old Age Home — operations support, resident assistance, facility upkeep.
 - Taught coding & digital literacy to underprivileged students (NGO).
 - Organized food/clothing drives for underserved communities.
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Achievement Highlights (Metrics-ready bullets)

- Reduced global stream latency by targeted **buffer/jitter** tuning and concurrent pipeline tweaks (RTSP/WebRTC path).
 - Drove **15+ ONVIF API** compatibility improvements → fewer interop failures with third-party NVR/VMS.
 - Delivered **OTA** pipeline supporting **1,000+ deployed devices** with rollback and health checks → lower field risk.
 - Consolidated common parsers/utils, cutting config-related defects and speeding team integration work across **7–9 engineers**.
 - Designed **30s local CVR buffer** to absorb WAN outages → fewer missed events, better customer experience.
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Role-focused Bullet Banks (Copy/Prune per JD)

Embedded Firmware Engineer

- C/C++ on embedded Linux for IP cameras; POSIX threads, epoll/select, timers, signals, IPC.
- Device mgmt: OTA updates (staged downloads, verification, rollback), persistent logs, watchdogs.
- File systems/SQLite3 tuning (WAL, indices); schema migrations for feature rollouts.
- Secure comms: TLS/digest (OpenSSL), credential lifecycle, token refresh.

Robotics/Perception Engineer

- ROS graph design; sensor fusion (camera+IMU/pressure); PID controllers; mission state machines.
- OpenCV pipelines (filtering, morphology, contouring, color spaces) under adverse underwater lighting.
- CUDA acceleration on Jetson for frame processing; profiling & kernel-level optimizations.

Systems/Kernel (OS) Engineer

- Implemented kernel **schedulers** (priority donation, CPU usage/load average), and timer-backed **sleep/wakeup**.
- Concurrency: locks, semaphores, condition variables; critical sections with interrupt mask windows.
- User programs: syscalls, stack arg passing, file I/O, exec/wait; page protection for executables.

Cloud/Edge Streaming Engineer

- Live555 RTSP/RTSPS media serving; WebRTC P2P/relay with Mosquitto signaling; buffering & backpressure.
- CVR: event-based recording; pre/post buffers; retry/backoff; token rotation; retention policies.
- Observability: stats hooks for RTT/loss/latency; reproducible field captures with Wireshark/tcpdump.

Data/ML Engineer (adjacent)

- E2E pipeline: cleaning, EDA, modeling; XGBoost/DT/SVM/MLP; clustering (K-Means).
 - Interpretable insights feeding product strategy (e.g., exercise moderates negative correlations of screen time).
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STAR Stories (Interview-ready)

- **ONVIF @ Scale** — *S*: Interop failures across NVRs due to spec-edge cases. *T*: Harden server & APIs. *A*: gSOAP+Lighttpd service with schema validation; refactored XML/JSON utilities; compliance tests. *R*: 15+ API issues resolved; smoother certifications; fewer field tickets.
 - **OTA Reliability** — *S*: Fleet updates bricked occasionally under power loss. *T*: Make updates resilient. *A*: Staged downloads, checksums, version gates, rollback, post-update health checks. *R*: Consistent rollouts to 1,000+ devices with minimal failures.
 - **Low-latency Streaming** — *S*: High E2E latency for remote viewers. *T*: Reduce latency without quality collapse. *A*: Tuning RTP pacing/buffers/jitter; WebRTC path with metrics; auth hardening. *R*: Lowered latency and improved stability in lossy conditions.
 - **CVR Resilience** — *S*: Event loss during WAN blips. *T*: Guarantee event capture. *A*: 30s local buffer, retry/backoff, digest/TLS, metadata indexing. *R*: Substantial drop in missed-event incidents.
 - **Pintos Priority Inversion** — *S*: High-priority thread blocked by low-priority lock holder. *T*: Implement donation. *A*: Priority donation with proper transitive propagation and revocation on unlock. *R*: Correctness under adversarial tests; deeper concurrency intuition.
 - **AUV CUDA Speedup** — *S*: CPU-bound vision caused sluggish control. *T*: Hit real-time FPS. *A*: CUDA accelerate filters and contours; pipeline restructure. *R*: ~15× speedup; stable control loops.
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Keyword Bank (ATS-friendly)

C, C++, Embedded Linux, POSIX, Multithreading, RTSP, RTSPS, RTP, WebRTC, Live555, FFmpeg, MQTT, Mosquitto, ONVIF G/S/T, gSOAP, Lighttpd, TLS, Digest Auth, OpenSSL, SQLite3, JSON, XML, cJSON, libxml2, ROS, OpenCV, CUDA, Jetson TX2, Raspberry Pi, STM32, SPI, I2C, UART, RS485, Jenkins, Docker, Wireshark, tcpdump, gdb, Valgrind, OTA, NVR, VMS, CVR, RBAC, Entitlements, Event Processing, NLP Search, Advanced Scheduler, Priority Donation, Alarm Clock, System Calls, File I/O, Process Management, FPGA, Verilog, Quartus, PID Control, EDA, XGBoost, SVM, Decision Trees, K-Means.

Optional Sections to Pull In (if JD needs)

- **Teaching/Mentorship:** Guided juniors on debugging with gdb/Valgrind; code reviews emphasizing thread safety and error handling patterns.
 - **Testing/QA:** Authored black-box test cases for ONVIF APIs; scripted scenario testers for OTA failure modes; packet-capture–driven regression checks.
 - **Security:** Rotating tokens for streaming endpoints; cipher suite restrictions; secure defaults; rate limiting and lockout policies.
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Quick Mini-summaries (for resume headers)

- **Embedded Firmware:** Built ONVIF-compliant camera firmware in C/C++ on Linux; OTA pipelines; WebRTC/RTSP streaming; TLS/digest auth; SQLite3-backed config at 1M+ user scale.
 - **Edge Streaming:** Designed low-latency RTSP/WebRTC pipelines and CVR with 30s local buffer; metrics for RTT/loss; robust under lossy networks.
 - **Robotics CV:** ROS + OpenCV on Jetson TX2 for AUV; PID control; CUDA-accelerated CV achieving real-time performance.
 - **Systems/OS:** Implemented Pintos schedulers (priority donation, advanced CPU/load), syscalls, and safe stack argument passing.
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Challenges faced: